School of Future Tech

Case Study Report

on

FitTrack Pro: Corporate Wellness

by

Pranav Santhosh Nair

150096725089

Index

1. Introduction to the Case Study.
2. Problem Statement / Case Background (Abstract).
3. Problem Statement / Case Study Design.
4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study.
5. Problem Statement / Case Study Implementation Details and Snapshots.
6. Problem Statement / Case Study Results and Conclusion.
7. Reference

1. Introduction to the Case Study

FitTrack Pro is a responsive web application that ITM Skills University created as a corporate wellness platform. The project shows how an interactive health-tracking interface that promotes employee engagement and activity can be created using contemporary web technologies.

- The program mimics a fitness monitoring setting in which users can:
- Keep track of your daily step totals.
- Track goal progress using visual cues
- Take part in fitness competitions
- View the standings on the leaderboard
- Record brief health updates

The application prioritises responsive design, visual feedback, and user engagement. FitTrack Pro is a lightweight, interactive platform that uses HTML5, CSS3, and JavaScript to encourage healthy habits through gamification features like milestones, badges, and progress visualisation.

2. Problem Statement / Case Background (Abstract)

Background

Corporate organizations are increasingly introducing wellness programs to improve employee health and productivity. However, many such platforms face problems like:

- Low employee engagement
- Lack of visual motivation
- Poor mobile responsiveness
- Limited team-based interaction
- Complex or heavy applications

Traditional fitness apps often focus only on individual tracking rather than team participation, which reduces motivation among employees.

Abstract

This case study introduces FitTrack Pro, a front-end corporate wellness dashboard built using HTML5, CSS3, and vanilla JavaScript. The application simulates a workplace wellness system where users can track steps, view progress toward fitness goals, and participate in wellness challenges.

The system includes several interactive components such as:

- Step tracking with real-time progress updates
- Gamification through milestone badges
- Leaderboard ranking
- Fitness challenges participation
- Responsive UI optimized for desktop and mobile

Users manually input steps, and the system dynamically updates the progress bar and milestone notifications. The project demonstrates how modern front-end technologies can create interactive and engaging user experiences without using heavy frameworks.

The case study evaluates the system based on:

- Usability
- Responsiveness
- Feature completeness
- Visual engagement

3. Case Study Design

User Interface Layout

The FitTrack Pro system is designed as a multi-page front-end prototype consisting of interconnected components:

- Home Page
- Dashboard
- Challenges Page
- Navigation Sidebar

Core Features

1. **Step Tracking System:-** Users can input their step counts manually, and the system updates the total step count dynamically.

2. **Progress Visualization:-** A dynamic progress bar displays the percentage of the daily goal completed.

3. **Gamification Elements:-** Milestones unlock motivational messages and badges such as:

- Halfway milestone
- Marathoner badge

4. **Challenge Participation:-** Users can join wellness challenges such as:

- 100k Steps Challenge
- Sleep Hygiene Challenge
- Hydration Challenge
- Lunchtime HIIT

5. **Interactive Dashboard:-** The dashboard page provides a visual summary of:

- Steps
- Calories
- Active minutes
- Sleep

Component Breakdown

1. **Navigation**

- Implemented using **CSS Flexbox**
- Collapsible sidebar for mobile screens
-

2. **Cards**

- Implemented using **CSS Grid**
- Adaptive layout automatically adjusts based on screen size

3. **Interactions**

JavaScript DOM event listeners manage:

- Step updates
- Progress calculations
- Hero image navigation

4. **Animations**

The UI includes:

- Fade-in animations
- Hover effects
- Progress bar transitions

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

1. **Step Accumulation Algorithm**

- Each time a user enters steps, they are added to the existing step count.

$$totalSteps=totalSteps+addedSteps$$

- Implemented in JavaScript:

```
totalSteps += addedSteps
```

2. **Progress Percentage Calculation**

- The system calculates how much of the daily goal is completed.

$$percentage=min(100,(totalSteps/GOAL)\times 100)$$

Where: GOAL = 10000 steps

This ensures progress does not exceed 100%.

3. Badge Awarding Logic

- The system uses **conditional statements** to trigger motivational messages.

Example logic:

```
if totalSteps >= 10000 → Marathoner badge
else if totalSteps >= 5000 → Halfway message
```

4. Hero Image Carousel Algorithm

- A rotating background image system is implemented using an **array and index counter**.

```
currentImgIndex = (currentImgIndex + 1) % images.length
```

- This ensures continuous looping of images.

5. Responsive Layout Algorithm

- The system uses **CSS Grid auto-fit functionality**:

```
grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
```

- This automatically adjusts the layout depending on screen width.

Technology Stack Used for the Case

1. Programming Languages

- **HTML5** – Structure of the application
- **CSS3** – Styling and responsive layout
  JavaScript (ES6) – Interactivity and DOM manipulation

2. Tools and Frameworks

- Vanilla JavaScript
- Font Awesome icons
  Unsplash images for hero backgrounds

3. Styling Technologies

- CSS Variables

```
--primary: #2ecc71
--secondary: #27ae60
```

- Flexbox layout

- CSS Grid layout
- Animations and transitions

Example:

```
@keyframes fadeIn
```

4. Deployment Type

The project is deployed as a **static website** consisting of:

- HTML files
- CSS stylesheets
- JavaScript files

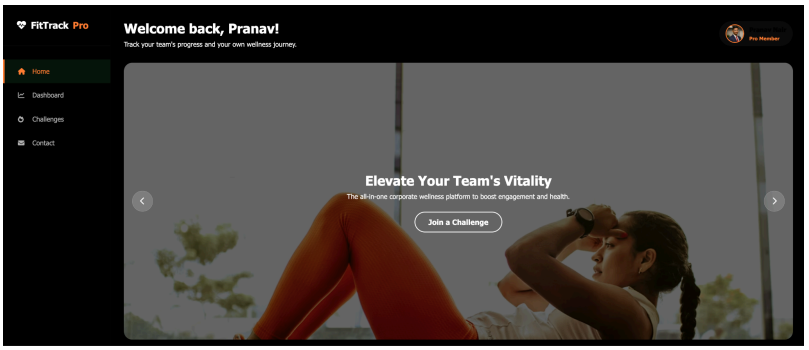
No backend or server processing is required.

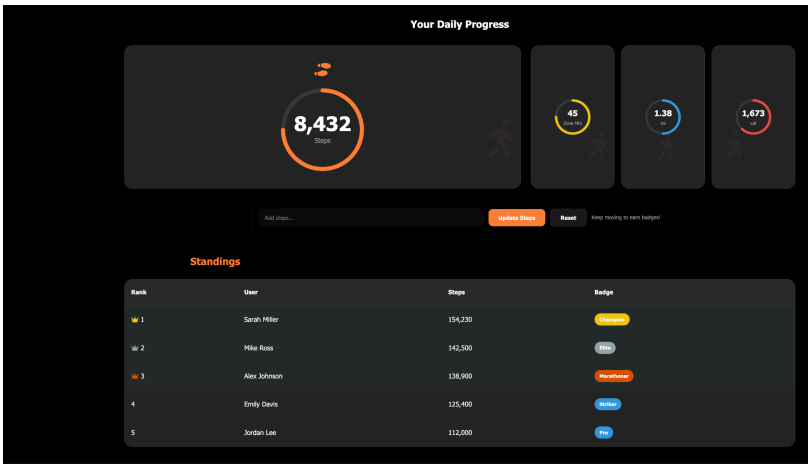
5. Application Interface

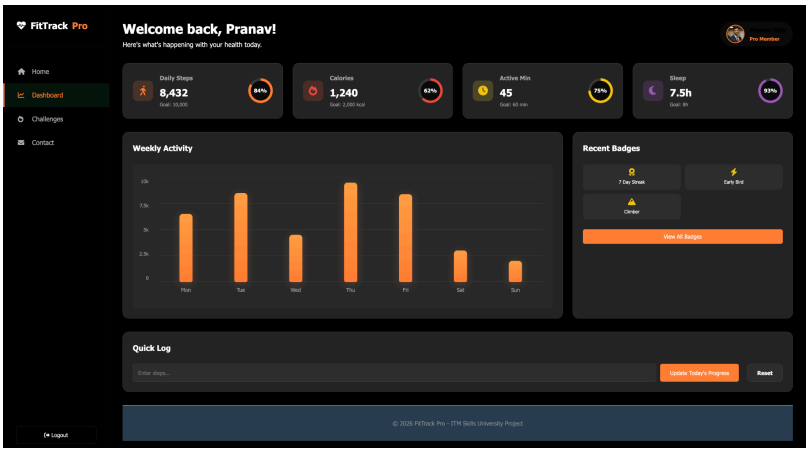
The interface contains:

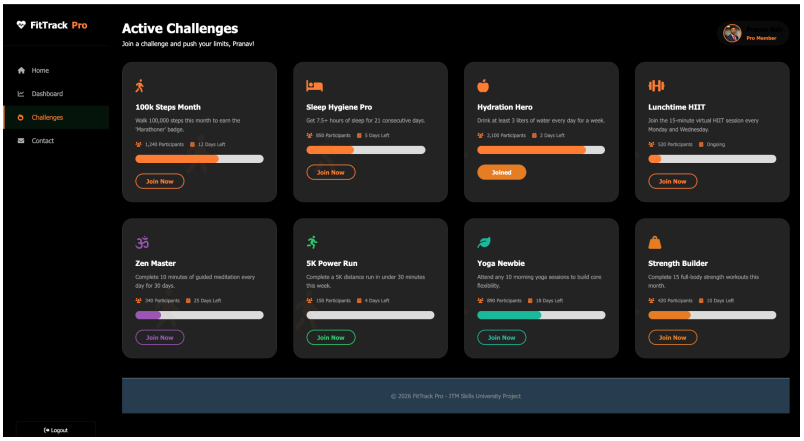
- Interactive cards
- Input fields for step logging
- Progress bars
- Leaderboard tables
- Responsive layouts
  These elements create a clean and user-friendly UI.

5. Problem Statement / Case Study Implementation Details and Snapshots.









6. Problem Statement / Case Study Results and Conclusion.

1. Findings

The results of the project demonstrate that:

- The step simulation system works smoothly across browsers.
- CSS Grid ensures fully responsive layouts.
- Interactive animations improve user engagement.
- Gamification elements such as badges increase motivation.
- The system achieves full functionality without external frameworks.

2. Conclusion

FitTrack Pro successfully demonstrates how core web technologies can be used to build an engaging wellness platform prototype.

The system provides:

- Interactive UI design
- Responsive layouts
- Gamified fitness tracking
- Lightweight performance

The project creates a strong foundation for future development, where additional features could include:

- Real-time fitness APIs
- User authentication
- Database integration
- Team competitions
- Mobile application versions

This makes FitTrack Pro an ideal academic case study for ITM Skills University projects focused on user-centered web application design.

7. References

- HTML5 Official Documentation – W3C
- CSS3 Documentation – MDN Web Docs
- JavaScript DOM API – Mozilla Developer Network
- Font Awesome Icons Library
- Unsplash Image Library for hero backgrounds